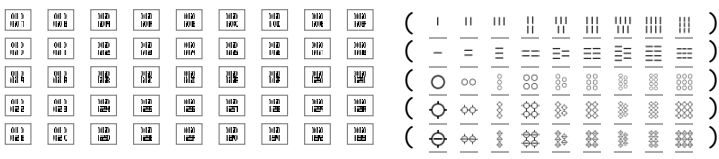
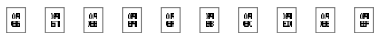


List of numeral systems

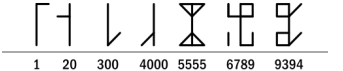
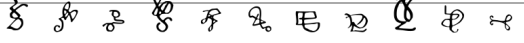
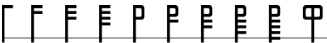
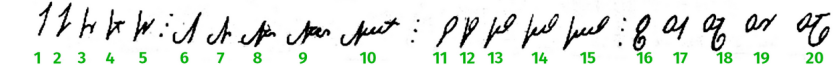
There are many different numeral systems, that is, writing systems for expressing numbers.

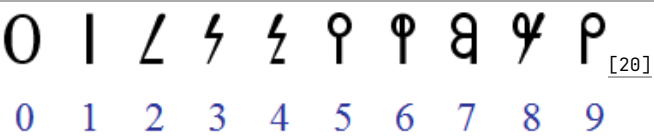
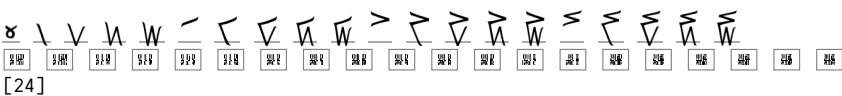
By culture / time period

"A *base* is a natural number B whose *powers* (B multiplied by itself some number of times) are specially designated within a numerical system."^[1]: 38 The term is not equivalent to radix, as it applies to all numerical notation systems (not just positional ones with a radix) and most systems of spoken numbers.^[1] Some systems have two bases, a smaller (subbase) and a larger (base); an example is Roman numerals, which are organized by fives ($V=5$, $L=50$, $D=500$, the subbase) and tens ($X=10$, $C=100$, $M=1,000$, the base).

Name	Base	Sample	Approx. First Appearance
Proto-cuneiform numerals	10&60		c. 3500–2000 BCE
Indus numerals	unknown ^[2]		c. 3500–1900 BCE ^[2]
Proto-Elamite numerals	10&60		3100 BCE
Sumerian numerals	10&60		3100 BCE
Egyptian numerals	10		3000 BCE
Babylonian numerals	10&60		2000 BCE
Aegean numerals	10		1500 BCE
Chinese numerals Japanese numerals Korean numerals (Sino-Korean) Vietnamese numerals (Sino-Vietnamese)	10	零一三四五六七八九十百千萬億 (Default, <u>Traditional Chinese</u>) 〇一二三四五六七八九十百千万亿 (Default, <u>Simplified Chinese</u>)	1300 BCE
Roman numerals	5&10	I V X L C D M	1000 BCE ^[1]
Hebrew numerals	10	ט ז ח טו ו ד ג ב א צ פ ע ס נ מ ל כ י ק ר ש ת ר מ ם ף ץ	800 BCE
Indian numerals	10	Bengali  Devanagari ० १ २ ३ ४ ५ ६ ७ ८ ९ Gujarati  Kannada  Malayalam  Odia  Punjabi  Tamil 	750–500 BCE

Name	Base	Sample	Approx. First Appearance
		<u>Telugu</u> ౦ ౧ ౨ ౩ ౪ ౫ ౬ ౭ ౮ ౯ <u>Tibetan</u> ༠ ༡ ༢ ༣ ༤ ༥ ༧ ༨ ༩ ༪ ༫ <u>Urdu</u> ۰ ۱ ۲ ۳ ۴ ۵ ۶ ۷ ۸ ۹	
<u>Greek numerals</u>	10	Ϟ α β γ δ ε Ϝ ζ η θ ι ο Α' Β' Γ' Δ' Ε' Ϛ' Ζ' Η' Θ'	<400 BCE
<u>Kharosthi numerals</u>	4&10	𑀓𑀔𑀕𑀖𑀗𑀘𑀙𑀚𑀛𑀜𑀝𑀞𑀟𑀠𑀡𑀢𑀣𑀤𑀥𑀦𑀧𑀨𑀩𑀪𑀫𑀬𑀭𑀮𑀯𑀰𑀱𑀲𑀳𑀴𑀵𑀶𑀷𑀸𑀹𑀺𑀻𑀼𑀽𑀾𑀿𑁀𑁁𑁂𑁃𑁄𑁅𑁆𑁇𑁈𑁉𑁊𑁋𑁌𑁍𑁎𑁏𑁐𑁑𑁒𑁓𑁔𑁕𑁖𑁗𑁘𑁙𑁚𑁛𑁜𑁝𑁞𑁟𑁠𑁡𑁢𑁣𑁤𑁥𑁦𑁧𑁨𑁩𑁪𑁫𑁬𑁭𑁮𑁯𑁰𑁱𑁲𑁳𑁴𑁵𑁶𑁷𑁸𑁹𑁺𑁻𑁼𑁽𑁾𑁿𑂀𑂁𑂂𑂃𑂄𑂅𑂆𑂇𑂈𑂉𑂊𑂋𑂌𑂍𑂎𑂏𑂐𑂑𑂒𑂓𑂔𑂕𑂖𑂗𑂘𑂙𑂚𑂛𑂜𑂝𑂞𑂟𑂠𑂡𑂢𑂣𑂤𑂥𑂦𑂧𑂨𑂩𑂪𑂫𑂬𑂭𑂮𑂯𑂰𑂱𑂲𑂳𑂴𑂵𑂶𑂷𑂸𑂺𑂹𑂻𑂼𑂽𑂾𑂿𑃀𑃁𑃂𑃃𑃄𑃅𑃆𑃇𑃈𑃉𑃊𑃋𑃌𑃍𑃎𑃏𑃐𑃑𑃒𑃓𑃔𑃕𑃖𑃗𑃘𑃙𑃚𑃛𑃜𑃝𑃞𑃟𑃠𑃡𑃢𑃣𑃤𑃥𑃦𑃧𑃨𑃩𑃪𑃫𑃬𑃭𑃮𑃯𑃰𑃱𑃲𑃳𑃴𑃵𑃶𑃷𑃸𑃹𑃺𑃻𑃼𑃽𑃾𑃿𑄀𑄁𑄂𑄃𑄄𑄅𑄆𑄇𑄈𑄉𑄊𑄋𑄌𑄍𑄎𑄏𑄐𑄑𑄒𑄓𑄔𑄕𑄖𑄗𑄘𑄙𑄚𑄛𑄜𑄝𑄞𑄟𑄠𑄡𑄢𑄣𑄤𑄥𑄦𑄧𑄨𑄩𑄪𑄫𑄬𑄭𑄮𑄯𑄰𑄱𑄲𑄳𑄴𑄵𑄶𑄷𑄸𑄹𑄺𑄻𑄼𑄽𑄾𑄿𑅀𑅁𑅂𑅃𑅄𑅅𑅆𑅇𑅈𑅉𑅊𑅋𑅌𑅍𑅎𑅏𑅐𑅑𑅒𑅓𑅔𑅕𑅖𑅗𑅘𑅙𑅚𑅛𑅜𑅝𑅞𑅟𑅠𑅡𑅢𑅣𑅤𑅥𑅦𑅧𑅨𑅩𑅪𑅫𑅬𑅭𑅮𑅯𑅰𑅱𑅲𑅳𑅴𑅵𑅶𑅷𑅸𑅹𑅺𑅻𑅼𑅽𑅾𑅿𑆀𑆁𑆂𑆃𑆄𑆅𑆆𑆇𑆈𑆉𑆊𑆋𑆌𑆍𑆎𑆏𑆐𑆑𑆒𑆓𑆔𑆕𑆖𑆗𑆘𑆙𑆚𑆛𑆜𑆝𑆞𑆟𑆠𑆡𑆢𑆣𑆤𑆥𑆦𑆧𑆨𑆩𑆪𑆫𑆬𑆭𑆮𑆯𑆰𑆱𑆲𑆳𑆴𑆵𑆶𑆷𑆸𑆹𑆺𑆻𑆼𑆽𑆾𑆿𑇀𑇁𑇂𑇃𑇄𑇅𑇆𑇇𑇈𑇉𑇊𑇋𑇌𑇍𑇎𑇏𑇐𑇑𑇒𑇓𑇔𑇕𑇖𑇗𑇘𑇙𑇚𑇛𑇜𑇝𑇞𑇟𑇠𑇡𑇢𑇣𑇤𑇥𑇦𑇧𑇨𑇩𑇪𑇫𑇬𑇭𑇮𑇯𑇰𑇱𑇲𑇳𑇴𑇵𑇶𑇷𑇸𑇹𑇺𑇻𑇼𑇽𑇾𑇿𑈀𑈁𑈂𑈃𑈄𑈅𑈆𑈇𑈈𑈉𑈊𑈋𑈌𑈍𑈎𑈏𑈐𑈑𑈒𑈓𑈔𑈕𑈖𑈗𑈘𑈙𑈚𑈛𑈜𑈝𑈞𑈟𑈠𑈡𑈢𑈣𑈤𑈥𑈦𑈧𑈨𑈩𑈪𑈫𑈬𑈭𑈮𑈯𑈰𑈱𑈲𑈳𑈴𑈶𑈵𑈷𑈸𑈹𑈺𑈻𑈼𑈽𑈾𑈿𑉀𑉁𑉂𑉃𑉄𑉅𑉆𑉇𑉈𑉉𑉊𑉋𑉌𑉍𑉎𑉏𑉐𑉑𑉒𑉓𑉔𑉕𑉖𑉗𑉘𑉙𑉚𑉛𑉜𑉝𑉞𑉟𑉠𑉡𑉢𑉣𑉤𑉥𑉦𑉧𑉨𑉩𑉪𑉫𑉬𑉭𑉮𑉯𑉰𑉱𑉲𑉳𑉴𑉵𑉶𑉷𑉸𑉹𑉺𑉻𑉼𑉽𑉾𑉿𑊀𑊁𑊂𑊃𑊄𑊅𑊆𑊇𑊈𑊉𑊊𑊋𑊌𑊍𑊎𑊏𑊐𑊑𑊒𑊓𑊔𑊕𑊖𑊗𑊘𑊙𑊚𑊛𑊜𑊝𑊞𑊟𑊠𑊡𑊢𑊣𑊤𑊥𑊦𑊧𑊨𑊩𑊪𑊫𑊬𑊭𑊮𑊯𑊰𑊱𑊲𑊳𑊴𑊵𑊶𑊷𑊸𑊹𑊺𑊻𑊼𑊽𑊾𑊿𑋀𑋁𑋂𑋃𑋄𑋅𑋆𑋇𑋈𑋉𑋊𑋋𑋌𑋍𑋎𑋏𑋐𑋑𑋒𑋓𑋔𑋕𑋖𑋗𑋘𑋙𑋚𑋛𑋜𑋝𑋞𑋟𑋠𑋡𑋢𑋣𑋤𑋥𑋦𑋧𑋨𑋩𑋪𑋫𑋬𑋭𑋮𑋯𑋰𑋱𑋲𑋳𑋴𑋵𑋶𑋷𑋸𑋹𑋺𑋻𑋼𑋽𑋾𑋿𑌀𑌁𑌂𑌃𑌄𑌅𑌆𑌇𑌈𑌉𑌊𑌋𑌌𑌍𑌎𑌏𑌐𑌑𑌒𑌓𑌔𑌕𑌖𑌗𑌘𑌙𑌚𑌛𑌜𑌝𑌞𑌟𑌠𑌡𑌢𑌣𑌤𑌥𑌦𑌧𑌨𑌩𑌪𑌫𑌬𑌭𑌮𑌯𑌰𑌱𑌲𑌳𑌴𑌵𑌶𑌷𑌸𑌹𑌺𑌻𑌼𑌽𑌾𑌿𑍀𑍁𑍂𑍃𑍄𑍅𑍆𑍇𑍈𑍉𑍊𑍋𑍌𑍍𑍎𑍏𑍐𑍑𑍒𑍓𑍔𑍕𑍖𑍗𑍘𑍙𑍚𑍛𑍜𑍝𑍞𑍟𑍠𑍡𑍢𑍣𑍤𑍥𑍦𑍧𑍨𑍩𑍪𑍫𑍬𑍭𑍮𑍯𑍰𑍱𑍲𑍳𑍴𑍵𑍶𑍷𑍸𑍹𑍺𑍻𑍼𑍽𑍾𑍿𑎀𑎁𑎂𑎃𑎄𑎅𑎆𑎇𑎈𑎉𑎊𑎋𑎌𑎍𑎎𑎏𑎐𑎑𑎒𑎓𑎔𑎕𑎖𑎗𑎘𑎙𑎚𑎛𑎜𑎝𑎞𑎟𑎠𑎡𑎢𑎣𑎤𑎥𑎦𑎧𑎨𑎩𑎪𑎫𑎬𑎭𑎮𑎯𑎰𑎱𑎲𑎳𑎴𑎵𑎶𑎷𑎸𑎹𑎺𑎻𑎼𑎽𑎾𑎿𑏀𑏁𑏂𑏃𑏄𑏅𑏆𑏇𑏈𑏉𑏊𑏋𑏌𑏍𑏎𑏏𑏐𑏑𑏒𑏓𑏔𑏕𑏖𑏗𑏘𑏙𑏚𑏛𑏜𑏝𑏟𑏠𑏡𑏢𑏣𑏤𑏥𑏦𑏧𑏨𑏩𑏪𑏫𑏬𑏭𑏮𑏯𑏰𑏱𑏲𑏳𑏴𑏵𑏶𑏷𑏸𑏹𑏺𑏻𑏼𑏽𑏾𑏿𑐀𑐁𑐂𑐃𑐄𑐅𑐆𑐇𑐈𑐉𑐊𑐋𑐌𑐍𑐎𑐏𑐐𑐑𑐒𑐓𑐔𑐕𑐖𑐗𑐘𑐙𑐚𑐛𑐜𑐝𑐞𑐟𑐠𑐡𑐢𑐣𑐤𑐥𑐦𑐧𑐨𑐩𑐪𑐫𑐬𑐭𑐮𑐯𑐰𑐱𑐲𑐳𑐴𑐵𑐶𑐷𑐸𑐹𑐺𑐻𑐼𑐽𑐾𑐿𑑀𑑁𑑂𑑃𑑄𑑅𑑆𑑇𑑈𑑉𑑊𑑋𑑌𑑍𑑎𑑏𑑐𑑑𑑒𑑓𑑔𑑕𑑖𑑗𑑘𑑙𑑚𑑛𑑜𑑝𑑞𑑟𑑠𑑡𑑢𑑣𑑤𑑥𑑦𑑧𑑨𑑩𑑪𑑫𑑬𑑭𑑮𑑯𑑰𑑱𑑲𑑳𑑴𑑵𑑶𑑷𑑸𑑹𑑺𑑻𑑼𑑽𑑾𑑿𑒀𑒁𑒂𑒃𑒄𑒅𑒆𑒇𑒈𑒉𑒊𑒋𑒌𑒍𑒎𑒏𑒐𑒑𑒒𑒓𑒔𑒕𑒖𑒗𑒘𑒙𑒚𑒛𑒜𑒝𑒞𑒟𑒠𑒡𑒢𑒣𑒤𑒥𑒦𑒧𑒨𑒩𑒪𑒫𑒬𑒭𑒮𑒯𑒰𑒱𑒲𑒳𑒴𑒵𑒶𑒷𑒸𑒻𑒻𑒼𑒽𑒾𑒿𑓀𑓁𑓃𑓂𑓄𑓅𑓆𑓇𑓈𑓉𑓊𑓋𑓌𑓍𑓎𑓏𑓐𑓑𑓒𑓓𑓔𑓕𑓖𑓗𑓘𑓙𑓚𑓛𑓜𑓝𑓞𑓟𑓠𑓡𑓢𑓣𑓤𑓥𑓦𑓧𑓨𑓩𑓪𑓫𑓬𑓭𑓮𑓯𑓰𑓱𑓲𑓳𑓴𑓵𑓶𑓷𑓸𑓹𑓺𑓻𑓼𑓽𑓾𑓿𑔀𑔁𑔂𑔃𑔄𑔅𑔆𑔇𑔈𑔉𑔊𑔋𑔌𑔍𑔎𑔏𑔐𑔑𑔒𑔓𑔔𑔕𑔖𑔗𑔘𑔙𑔚𑔛𑔜𑔝𑔞𑔟𑔠𑔡𑔢𑔣𑔤𑔥𑔦𑔧𑔨𑔩𑔪𑔫𑔬𑔭𑔮𑔯𑔰𑔱𑔲𑔳𑔴𑔵𑔶𑔷𑔸𑔹𑔺𑔻𑔼𑔽𑔾𑔿𑕀𑕁𑕂𑕃𑕄𑕅𑕆𑕇𑕈𑕉𑕊𑕋𑕌𑕍𑕎𑕏𑕐𑕑𑕒𑕓𑕔𑕕𑕖𑕗𑕘𑕙𑕚𑕛𑕜𑕝𑕞𑕟𑕠𑕡𑕢𑕣𑕤𑕥𑕦𑕧𑕨𑕩𑕪𑕫𑕬𑕭𑕮𑕯𑕰𑕱𑕲𑕳𑕴𑕵𑕶𑕷𑕸𑕹𑕺𑕻𑕼𑕽𑕾𑕿𑖀𑖁𑖂𑖃𑖄𑖅𑖆𑖇𑖈𑖉𑖊𑖋𑖌𑖍𑖎𑖏𑖐𑖑𑖒𑖓𑖔𑖕𑖖𑖗𑖘𑖙𑖚𑖛𑖜𑖝𑖞𑖟𑖠𑖡𑖢𑖣𑖤𑖥𑖦𑖧𑖨𑖩𑖪𑖫𑖬𑖭𑖮𑖯𑖰𑖱𑖲𑖳𑖴𑖵𑖶𑖷𑖸𑖹𑖺𑖻𑖼𑖽𑖾𑗀𑖿𑗁𑗂𑗃𑗄𑗅𑗆𑗇𑗈𑗉𑗊𑗋𑗌𑗍𑗎𑗏𑗐𑗑𑗒𑗓𑗔𑗕𑗖𑗗𑗘𑗙𑗚𑗛𑗜𑗝𑗞𑗟𑗠𑗡𑗢𑗣𑗤𑗥𑗦𑗧𑗨𑗩𑗪𑗫𑗬𑗭𑗮𑗯𑗰𑗱𑗲𑗳𑗴𑗵𑗶𑗷𑗸𑗹𑗺𑗻𑗼𑗽𑗾𑗿𑘀𑘁𑘂𑘃𑘄𑘅𑘆𑘇𑘈𑘉𑘊𑘋𑘌𑘍𑘎𑘏𑘐𑘑𑘒𑘓𑘔𑘕𑘖𑘗𑘘𑘙𑘚𑘛𑘜𑘝𑘞𑘟𑘠𑘡𑘢𑘣𑘤𑘥𑘦𑘧𑘨𑘩𑘪𑘫𑘬𑘭𑘮𑘯𑘰𑘱𑘲𑘳𑘴𑘵𑘶𑘷𑘸𑘹𑘺𑘻𑘼𑘽𑘾𑘿𑙀𑙁𑙂𑙃𑙄𑙅𑙆𑙇𑙈𑙉𑙊𑙋𑙌𑙍𑙎𑙏𑙐𑙑𑙒𑙓𑙔𑙕𑙖𑙗𑙘𑙙𑙚𑙛𑙜𑙝𑙞𑙟𑙠𑙡𑙢𑙣𑙤𑙥𑙦𑙧𑙨𑙩𑙪𑙫𑙬𑙭𑙮𑙯𑙰𑙱𑙲𑙳𑙴𑙵𑙶𑙷𑙸𑙹𑙺𑙻𑙼𑙽𑙾𑙿𑚀𑚁𑚂𑚃𑚄𑚅𑚆𑚇𑚈𑚉𑚊𑚋𑚌𑚍𑚎𑚏𑚐𑚑𑚒𑚓𑚔𑚕𑚖𑚗𑚘𑚙𑚚𑚛𑚜𑚝𑚞𑚟𑚠𑚡𑚢𑚣𑚤𑚥𑚦𑚧𑚨𑚩𑚪𑚫𑚬𑚭𑚮𑚯𑚰𑚱𑚲𑚳𑚴𑚵𑚷𑚶𑚸𑚹𑚺𑚻𑚼𑚽𑚾𑚿𑛀𑛁𑛂𑛃𑛄𑛅𑛆𑛇𑛈𑛉𑛊𑛋𑛌𑛍𑛎𑛏𑛐𑛑𑛒𑛓𑛔𑛕𑛖𑛗𑛘𑛙𑛚𑛛𑛜𑛝𑛞𑛟𑛠𑛡𑛢𑛣𑛤𑛥𑛦𑛧𑛨𑛩𑛪𑛫𑛬𑛭𑛮𑛯𑛰𑛱𑛲𑛳𑛴𑛵𑛶𑛷𑛸𑛹𑛺𑛻𑛼𑛽𑛾𑛿𑜀𑜁𑜂𑜃𑜄𑜅𑜆𑜇𑜈𑜉𑜊𑜋𑜌𑜍𑜎𑜏𑜐𑜑𑜒𑜓𑜔𑜕𑜖𑜗𑜘𑜙𑜚𑜛𑜜𑜝𑜞𑜟𑜠𑜡𑜢𑜣𑜤𑜥𑜦𑜧𑜨𑜩𑜪𑜫𑜬𑜭𑜮𑜯𑜰𑜱𑜲𑜳𑜴𑜵𑜶𑜷𑜸𑜹𑜺𑜻𑜼𑜽𑜾𑜿𑝀𑝁𑝂𑝃𑝄𑝅𑝆𑝇𑝈𑝉𑝊𑝋𑝌𑝍𑝎𑝏𑝐𑝑𑝒𑝓𑝔𑝕𑝖𑝗𑝘𑝙𑝚𑝛𑝜𑝝𑝞𑝟𑝠𑝡𑝢𑝣𑝤𑝥𑝦𑝧𑝨𑝩𑝪𑝫𑝬𑝭𑝮𑝯𑝰𑝱𑝲𑝳𑝴𑝵𑝶𑝷𑝸𑝹𑝺𑝻𑝼𑝽𑝾𑝿𑞀𑞁𑞂𑞃𑞄𑞅𑞆𑞇𑞈𑞉𑞊𑞋𑞌𑞍𑞎𑞏𑞐𑞑𑞒𑞓𑞔𑞕𑞖𑞗𑞘𑞙𑞚𑞛𑞜𑞝𑞞𑞟𑞠𑞡𑞢𑞣𑞤𑞥𑞦𑞧𑞨𑞩𑞪𑞫𑞬𑞭𑞮𑞯𑞰𑞱𑞲𑞳𑞴𑞵𑞶𑞷𑞸𑞹𑞺𑞻𑞼𑞽𑞾𑞿𑟀𑟁𑟂𑟃𑟄𑟅𑟆𑟇𑟈𑟉𑟊𑟋𑟌𑟍𑟎𑟏𑟐𑟑𑟒𑟓𑟔𑟕𑟖𑟗𑟘𑟙𑟚𑟛𑟜𑟝𑟞𑟟𑟠𑟡𑟢𑟣𑟤𑟥𑟦𑟧𑟨𑟩𑟪𑟫𑟬𑟭𑟮𑟯𑟰𑟱𑟲𑟳𑟴𑟵𑟶𑟷𑟸𑟹𑟺𑟻𑟼𑟽𑟾𑟿𑠀𑠁𑠂𑠃𑠄𑠅𑠆𑠇𑠈𑠉𑠊𑠋𑠌𑠍𑠎𑠏𑠐𑠑𑠒𑠓𑠔𑠕𑠖𑠗𑠘𑠙𑠚𑠛𑠜𑠝𑠞𑠟𑠠𑠡𑠢𑠣𑠤𑠥𑠦𑠧𑠨𑠩𑠪𑠫𑠬𑠭𑠮𑠯𑠰𑠱𑠲𑠳𑠴𑠵𑠶𑠷𑠸𑠺𑠹𑠻𑠼𑠽𑠾𑠿𑡀𑡁𑡂𑡃𑡄𑡅𑡆𑡇𑡈𑡉𑡊𑡋𑡌𑡍𑡎𑡏𑡐𑡑𑡒𑡓𑡔𑡕𑡖𑡗𑡘𑡙𑡚𑡛𑡜𑡝𑡞𑡟𑡠𑡡𑡢𑡣𑡤𑡥𑡦𑡧𑡨𑡩𑡪𑡫𑡬𑡭𑡮𑡯𑡰𑡱𑡲𑡳𑡴𑡵𑡶𑡷𑡸𑡹𑡺𑡻𑡼𑡽𑡾𑡿𑢀𑢁𑢂𑢃𑢄𑢅𑢆𑢇𑢈𑢉𑢊𑢋𑢌𑢍𑢎𑢏𑢐𑢑𑢒𑢓𑢔𑢕𑢖𑢗𑢘𑢙𑢚𑢛𑢜𑢝𑢞𑢟𑢠𑢡𑢢𑢣𑢤𑢥𑢦𑢧𑢨𑢩𑢪𑢫𑢬𑢭𑢮𑢯𑢰𑢱𑢲𑢳𑢴𑢵𑢶𑢷𑢸𑢹𑢺𑢻𑢼𑢽𑢾𑢿𑣀𑣁𑣂𑣃𑣄𑣅𑣆𑣇𑣈𑣉𑣊𑣋𑣌𑣍𑣎𑣏𑣐𑣑𑣒𑣓𑣔𑣕𑣖𑣗𑣘𑣙𑣚𑣛𑣜𑣝𑣞𑣟𑣠𑣡𑣢𑣣𑣤𑣥𑣦𑣧𑣨𑣩𑣪𑣫𑣬𑣭𑣮𑣯𑣰𑣱𑣲𑣳𑣴𑣵𑣶𑣷𑣸𑣹𑣺𑣻𑣼𑣽𑣾𑣿𑤀𑤁𑤂𑤃𑤄𑤅𑤆𑤇𑤈𑤉𑤊𑤋𑤌𑤍𑤎𑤏𑤐𑤑𑤒𑤓𑤔𑤕𑤖𑤗𑤘𑤙𑤚𑤛𑤜𑤝𑤞𑤟𑤠𑤡𑤢𑤣𑤤𑤥𑤦𑤧𑤨𑤩𑤪𑤫𑤬𑤭𑤮𑤯𑤰𑤱𑤲𑤳𑤴𑤵𑤶𑤷𑤸𑤹𑤺𑤻𑤼𑤽𑤾𑤿𑥀𑥁𑥂𑥃𑥄𑥅𑥆𑥇𑥈𑥉𑥊𑥋𑥌𑥍𑥎𑥏𑥐𑥑𑥒𑥓𑥔𑥕𑥖𑥗𑥘𑥙𑥚𑥛𑥜𑥝𑥞𑥟𑥠𑥡𑥢𑥣𑥤𑥥𑥦𑥧𑥨𑥩𑥪𑥫𑥬𑥭𑥮𑥯	

Name	Base	Sample	Approx. First Appearance
<u>Burmese numerals</u>	10		11th century ^[10]
<u>Tangut numerals</u>	10		11th century (1036)
<u>Cistercian numerals</u>	10	 1 20 300 4000 5555 6789 9394	13th century
<u>Maya numerals</u>	5&20		15th century
<u>Muisca numerals</u>	20	ATA BOSA MICA MUHICA HISCA TA CUHUPCUA SUHUSA ACA UBCHHICA GUETA Acosta  Humboldt  Zerda  1 2 3 4 5 6 7 8 9 10 20	15th century
<u>Korean numerals (Hangul)</u>	10	영 일 이 삼 사 오 육 칠 팔 구	15th century (1443)
<u>Aztec numerals</u>	20	 (1, 5, 20, 100, 400, 800, 8000)	16th century
<u>Sinhala numerals</u>	10		18th century
<u>Pentadic runes</u>	10		19th century
<u>Cherokee numerals</u>	10	 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20	19th century (1820s)
<u>Vai numerals</u>	10	 [11]	19th century (1832) ^[12]
<u>Bamum numerals</u>	10	 [13]	19th century (1896) ^[12]
<u>Mende Kikakui numerals</u>	10	 [14]	20th century (1917) ^[15]
<u>Osmanya numerals</u>	10		20th century (1920s)
<u>Medefaidrin numerals</u>	20	 [16]	20th century (1930s) ^[17]
<u>N'Ko numerals</u>	10	 [18]	20th century (1949) ^[19]

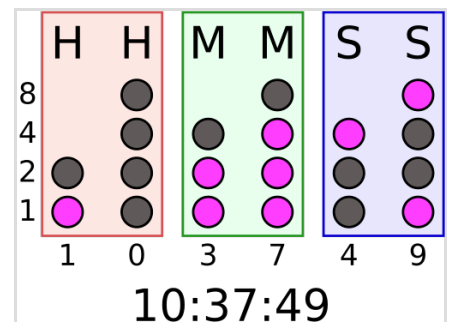
Name	Base	Sample	Approx. First Appearance
<u>Hmong numerals</u>	10		20th century (1959)
<u>Garay numerals</u>	10	 ^[20]	20th century (1961) ^[21]
<u>Adlam numerals</u>	10	 ^[22]	20th century (1989) ^[23]
<u>Kaktovik numerals</u>	5&20	 ^[24]	20th century (1994) ^[25]
<u>Sundanese numerals</u>	10		20th century (1996) ^[26]

By type of notation

Numeral systems are classified here as to whether they use positional notation (also known as place-value notation), and further categorized by radix or base.

Standard positional numeral systems

The common names are derived somewhat arbitrarily from a mix of Latin and Greek, in some cases including roots from both languages within a single name.^[27] There have been some proposals for standardisation.^[28]



A binary clock might use LEDs to express binary values. In this clock, each column of LEDs shows a binary-coded decimal numeral of the traditional sexagesimal time.

Base	Name	Usage
2	<u>Binary</u>	<u>Digital computing</u> , <u>imperial</u> and <u>customary volume</u> (<u>bushel-kenning-peck-gallon-pottle-quart-pint-cup-gill-jack-fluid ounce-tablespoon</u>).
3	<u>Ternary</u> , <u>trinary</u> ^[29]	Cantor set (all points in [0,1] that <i>can</i> be represented in ternary with no 1s); <u>counting Tasbih</u> in Islam; <u>hand-foot-yard</u> and <u>teaspoon-tablespoon-shot measurement systems</u> ; most <u>economical</u> integer base.
4	<u>Quaternary</u>	<u>Chumashan languages</u> and <u>Kharosthi numerals</u> .
5	<u>Quinary</u>	<u>Aneityum</u> (traditional), ^[30] <u>Ateso</u> , <u>Gumatj</u> , <u>Kuurn Kopan Noot</u> , and <u>Nunggubuyu</u> , <u>Saraveca languages</u> ; common count grouping e.g. <u>tally marks</u> .
6	<u>Senary</u> , <u>seximal</u>	<u>Diceware</u> , <u>Ndom</u> , <u>Kanum</u> , and <u>Proto-Uralic language</u> (suspected).
7	<u>Septenary</u> , <u>septimal</u>	
8	<u>Octal</u>	<u>Charles XII of Sweden</u> , <u>Unix-like permissions</u> , <u>Squawk codes</u> , <u>DEC PDP-11</u> , <u>Yuki</u> , <u>Pame</u> , compact notation for binary numbers, <u>Xiantian</u> (<u>I Ching</u> , <u>China</u>).
9	<u>Nonary</u> , <u>nonal</u>	Compact notation for <u>ternary</u> .
10	<u>Decimal</u> , <u>denary</u>	Most widely used by contemporary societies. ^{[31][32][33]}
11	<u>Undecimal</u> , <u>unodecimal</u> , <u>undenary</u>	A base-11 number system was mistakenly attributed to the Māori (New Zealand) in the 19th century ^[34] and one was reported to be used by the Pangwa (Tanzania) in the 20th century, ^[35] but was not confirmed by later research and is believed to also be an error. ^[36] Briefly proposed during the French Revolution to settle a dispute between those proposing a shift to duodecimal and those who were content with decimal. Used as a check digit in ISBN for 10-digit ISBNs. Applications in computer science and technology. ^{[37][38][39]}
12	<u>Duodecimal</u> , <u>dozenal</u>	Languages in the Nigerian Middle Belt Janji, <u>Gbiri-Niragu</u> , <u>Piti</u> , and the Nimbia dialect of <u>Gwandara</u> ; <u>Chepeng language</u> of <u>Nepal</u> , and the Mahl dialect of <u>Maldivian</u> ; <u>dozen-gross-great gross counting</u> ; <u>12-hour clock</u> and <u>months timekeeping</u> ; years of <u>Chinese zodiac</u> ; <u>foot</u> and <u>inch</u> ; <u>Roman fractions</u> .
13	<u>Tredecimal</u> , <u>tridecimal</u> ^{[40][41]}	<u>Conway's base 13 function</u> .
14	<u>Quattuordecimal</u> , <u>quadrodecimal</u> ^{[40][41]}	Programming for the <u>HP 9100A/B calculator</u> ^[42] and image processing applications. ^[43]
15	<u>Quindecimal</u> , <u>pentadecimal</u> ^{[44][41]}	Telephony routing over IP, and the <u>Huli language</u> . ^[36]
16	<u>Hexadecimal</u> , <u>sexadecimal</u> , <u>sedecimal</u>	Compact notation for <u>binary data</u> ; <u>tonal system</u> of <u>Nystrom</u> .
17	<u>Heptadecimal</u> , <u>septendecimal</u> ^{[44][41]}	
18	<u>Octodecimal</u> ^{[44][41]}	
19	<u>Undevicesimal</u> , <u>nonadecimal</u> ^{[44][41]}	
20	<u>Vigesimal</u>	<u>Basque</u> , <u>Celtic</u> , <u>Muisca</u> , <u>Inuit</u> , <u>Yoruba</u> , <u>Tlingit</u> , and <u>Dzongkha numerals</u> ; <u>Santali</u> , and <u>Ainu languages</u> .
5&20	<u>Quinary-vigesimal</u> ^[45] ^{[46][47]}	<u>Greenlandic</u> , <u>Iñupiaq</u> , <u>Kaktovik</u> , <u>Maya</u> , <u>Nunivak Cup'ig</u> , and <u>Yup'ik numerals</u> – "wide-spread... in the whole territory from Alaska along the Pacific Coast to the Orinoco and the Amazon." ^[45]
21		The smallest base in which all fractions $\frac{1}{2}$ to $\frac{1}{18}$ have <u>periods</u> of 4 or shorter.

23		<u>Kalam language</u> , ^[48]
24	<u>Quadragigesimal</u> ^[49]	<u>24-hour clock</u> timekeeping; <u>Greek alphabet</u> ; <u>Kaugel language</u> .
25		Compact notation for quinary.
26	<u>Hexavigesimal</u> ^{[49][50]}	Sometimes used for encryption or ciphering, ^[51] using all letters in the <u>English alphabet</u> . Used to encode <u>SHA-256</u> hashes into uppercase letters in <u>InChIKey</u> (a standard indexing system of chemical structures) ^[52] and <u>SID</u> (sequence identification, an indexing system of PCR amplicons in forensics). ^[50]
27		<u>Telefol</u> , ^[48] <u>Oksapmin</u> , ^[53] <u>Wambon</u> , ^[54] and <u>Hewa</u> ^[55] languages. Mapping the nonzero digits to the alphabet and zero to the space is occasionally used to provide <u>checksums</u> for alphabetic data such as personal names, ^[56] to provide a concise encoding of alphabetic strings, ^[57] or as the basis for a form of <u>gematria</u> . ^[58] Compact notation for <u>ternary</u> .
30	<u>Trigesimal</u> ^[59]	The <u>Natural Area Code</u> , this is the smallest base such that all of $\frac{1}{2}$ to $\frac{1}{6}$ terminate, a number n is a <u>regular number</u> if and only if $\frac{1}{n}$ terminates in base 30.
32	<u>Duotrigesimal</u>	Found in the <u>Ngiti language</u> . Also used to encode <u>computer (binary) data</u> into an alphanumerical string without <u>confusable characters</u> (e.g. zero and "0", eight and "B") in <u>RFC 4648</u> (https://www.rfc-editor.org/rfc/rfc4648), with each character standing for 5 bits.
34		The smallest base where $\frac{1}{2}$ terminates and all of $\frac{1}{2}$ to $\frac{1}{18}$ have periods of 4 or shorter.
36	<u>Hexatrigesimal</u> ^{[60][61]}	Used to encode large numbers into an alphanumeric string (26 letters, 10 numbers). Compact notation for <u>senary</u> .
40		<u>DEC RADIX 50/MOD40</u> encoding used to compactly represent file names and other symbols on <u>Digital Equipment Corporation</u> computers. The character set is a subset of <u>ASCII</u> consisting of space, upper case letters, the punctuation marks "\$", ".", and "%", and the numerals.
42		Largest base for which all <u>minimal primes</u> are known.
47		Smallest base for which no <u>generalized Wieferich primes</u> are known.
50		<u>SQUOZE</u> encoding used to compactly represent file names and other symbols on some <u>IBM</u> computers. Encoding using all <u>Gurmukhi</u> characters plus the <u>Gurmukhi</u> digits.
60	<u>Sexagesimal</u>	<u>Babylonian numerals</u> and <u>Sumerian</u> ; <u>degrees-minutes-seconds</u> and <u>hours-minutes-seconds</u> measurement systems; <u>Ekari</u> ; covers base 62 apart from <u>I</u> , <u>0</u> , and <u>l</u> , but including <u>_</u> (underscore). ^[62]
64		Used to encode <u>computer (binary) data</u> into a relatively compact string, with each character standing for 6 bits (<u>RFC 4648</u> (https://www.rfc-editor.org/rfc/rfc4648))).
72		The smallest base greater than binary such that no <u>three-digit narcissistic number</u> exists.
80		Used as a sub-base in <u>Supyire</u> .
89		Largest base for which all <u>left-truncatable primes</u> are known.
90		Related to <u>Goormaghtigh conjecture</u> for the generalized <u>repunit</u> numbers (111 in base 90 = 111111111111 in base 2).
97		Smallest base which is not perfect odd power (where generalized <u>Wagstaff numbers</u> can be factored algebraically) for which no generalized <u>Wagstaff primes</u> are known.

185		Smallest base which is not a <u>perfect power</u> (where generalized repunits can be factored algebraically) for which no generalized <u>repunit primes</u> are known.
210		Smallest base such that all fractions $\frac{1}{2}$ to $\frac{1}{10}$ terminate.

Non-standard positional numeral systems

Bijjective numeration

Base	Name	Usage
1	<u>Unary</u> (Bijjective base-1)	<u>Tally marks</u> , <u>Counting</u> . <u>Unary numbering</u> is used as part of some data compression algorithms such as <u>Golomb coding</u> . It also forms the basis for the <u>Peano axioms</u> for <u>formalizing arithmetic</u> within <u>mathematical logic</u> . A form of unary notation called <u>Church encoding</u> is used to represent numbers within <u>lambda calculus</u> . Some <u>email spam filters</u> tag messages with a number of asterisks in an e-mail header such as <i>X-Spam-Bar</i> or <i>X-SPAM-LEVEL</i> . The larger the number, the more likely the email is considered spam.
2	<u>Bijjective base-2</u>	
3	<u>Bijjective base-3</u>	
4	<u>Bijjective base-4</u>	
5	<u>Bijjective base-5</u>	
6	<u>Bijjective base-6</u>	
8	<u>Bijjective base-8</u>	
10	<u>Bijjective base-10</u>	To avoid <u>zero</u>
12	<u>Bijjective base-12</u>	
16	<u>Bijjective base-16</u>	
26	<u>Bijjective base-26</u>	<u>Spreadsheet column numeration</u> . Also used by John Nash as part of his obsession with <u>numerology</u> and the uncovering of "hidden" messages. [63]

Signed-digit representation

For odd bases b , the used digits are $-(b-1)/2$ to $(b-1)/2$, and for even bases b , the used digits are $-b/2$ to $b/2$, but for even bases b , the digits $-b/2$ and $b/2$ may not be followed by an odd digit.

Base	Name	Usage
2	Balanced binary (Non-adjacent form)	
3	Balanced ternary	<u>Ternary computers</u>
4	Balanced quaternary	
5	Balanced quinary	
6	Balanced senary	
7	Balanced septenary	
8	Balaced octal	
9	Balaced nonary	
10	Balanced decimal	John Colson <u>Augustin Cauchy</u>
11	Balaced undecimal	
12	Balaced duodecimal	

Negative bases

A negative base can write all (positive or negative or 0) integers without the negative sign. The common names of the negative base numeral systems are formed using the prefix *nega-*, giving names such as:

Base	Name	Usage
−2	Negabinary	
−3	Negaternary	
−4	Negaquaternary	
−5	Negaquinary	
−6	Negasenary	
−8	Negaoctal	
−10	Negadecimal	
−12	Negaduodecimal	
−16	Negahexadecimal	

Complex bases

Base	Name	Usage
$2i$	<u>Quater-imaginary base</u>	related to base −4 and base 16
$\sqrt{2}i$	Base $\sqrt{2}i$	related to base −2 and base 4
$\sqrt[4]{2}i$	Base $\sqrt[4]{2}i$	related to base 2
2ω	Base 2ω	related to base 8
$\sqrt[3]{2}\omega$	Base $\sqrt[3]{2}\omega$	related to base 2
$-1 \pm i$	<u>Twindragon</u> base	<u>Twindragon</u> fractal shape, related to base −4 and base 16
$1 \pm i$	Nega-Twindragon base	related to base −4 and base 16

Non-integer bases

Base	Name	Usage
$\frac{3}{2}$	Base $\frac{3}{2}$	a rational non-integer base
$\frac{4}{3}$	Base $\frac{4}{3}$	related to duodecimal
$\frac{5}{2}$	Base $\frac{5}{2}$	related to decimal
$\sqrt{2}$	Base $\sqrt{2}$	related to base 2
$\sqrt{3}$	Base $\sqrt{3}$	related to base 3
$\sqrt[3]{2}$	Base $\sqrt[3]{2}$	
$\sqrt[4]{2}$	Base $\sqrt[4]{2}$	
$\sqrt[12]{2}$	Base $\sqrt[12]{2}$	usage in 12-tone equal temperament musical system
$2\sqrt{2}$	Base $2\sqrt{2}$	
$-\frac{3}{2}$	Base $-\frac{3}{2}$	a negative rational non-integer base
$-\sqrt{2}$	Base $-\sqrt{2}$	a negative non-integer base, related to base 2
$\sqrt{10}$	Base $\sqrt{10}$	related to decimal
$2\sqrt{3}$	Base $2\sqrt{3}$	related to duodecimal
φ	Golden ratio base	Early Beta encoder ^[64]
ρ	Plastic ratio base	
ψ	Supergolden ratio base	
$1 + \sqrt{2}$	Silver ratio base	
e	Base e	Lowest radix economy
π	Base π	
$e\pi$	Base $e\pi$	
e^π	Base e^π	

n -adic number

n -adic number is a [field](#) if and only if n is [prime power](#).

Base	Name	Usage
2	Dyadic number	
3	Triadic number	
4	Tetradic number	the same as dyadic number
5	Pentadic number	
6	Hexadic number	not a field
7	Heptadic number	
8	Octadic number	the same as dyadic number
9	Enneadic number	the same as triadic number
10	Decadic number	not a field
11	Hendecadic number	
12	Dodecadic number	not a field

Mixed radix

- Factorial number system {1, 2, 3, 4, 5, 6, ...}
- Even double factorial number system {2, 4, 6, 8, 10, 12, ...}
- Odd double factorial number system {1, 3, 5, 7, 9, 11, ...}
- Primorial number system {2, 3, 5, 7, 11, 13, ...}
- Compositorial number system {4, 6, 8, 9, 10, 12, ...}
- Fibonorial number system {1, 2, 3, 5, 8, 13, ...}
- {60, 60, 24, 7} in timekeeping
- {60, 60, 24, 30 (or 31 or 28 or 29), 12, 10, 10, 10} in timekeeping
- (12, 20) traditional English monetary system (£sd)
- (20, 18, 13) Maya timekeeping

Other

- Quote notation
- Redundant binary representation
- Hereditary base-n notation
- Asymmetric numeral systems optimized for non-uniform probability distribution of symbols
- Combinatorial number system

Non-positional notation

All known numeral systems developed before the Babylonian numerals are non-positional,^[65] as are many developed later, such as the Roman numerals. The French Cistercian monks created their own numeral system.

See also

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-
- History of ancient numeral systems
 - History of the Hindu–Arabic numeral system
 - List of books on history of number systems

- List of numeral system topics
- Numeral prefix – Prefix derived from numerals or other numbers
- Radix – Number of digits of a numeral system
- Radix economy – Number of digits needed to express a number in a particular base
- Timeline of numerals and arithmetic

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